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| **Status** | Finished |
| **Started** | Monday, 26 October 2026, 1:27 AM |
| **Completed** | Monday, 26 October 2026, 1:30 AM |
| **Duration** | 3 mins 5 secs |
| **Grade** | **13.00** out of 15.00 (**86.67**%) |

Question **1**

Incorrect

Mark 0.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists
b. Binary search does not require dividing the list ✗
c. Binary search can find multiple instances of the target element
d. Binary search works on unsorted lists

Question **2**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops
b. The higher middle element is chosen as the middle element
c. The middle element is chosen randomly
d. The lower middle element is chosen as the middle element ✗

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. An error is raised
c. The first element is returned
d. The search is considered unsuccessful ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Checking each element sequentially ✓
c. Skipping every second element
d. Dividing the list in half repeatedly

Question **5**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Sequential search
c. Binary search ✓
d. Linear search

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list contains duplicate elements
c. When the list is unsorted
d. When the list is sorted ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list ✓
b. Divide the list into two equal parts
c. Sort the list
d. Compare the target element with the first element in the list

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **9**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O🥐
c. O(1) ✓
d. O(log n)

Question **10**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear
b. Binary ✓
c. Hash
d. Both (1) & (3)

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Linear search ✓
b. Merge search
c. Quick search
d. Bubble search

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. n ✓
c. log n
d. 1

Question **13**

Correct

Mark 1.00 out of 1.00

**_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.**

a. Insertion
b. Bubble ✓
c. Selection
d. Complexity

Question **14**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search continues in the left sublist
c. The search continues in the right sublist
d. The search ends successfully ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search stops immediately ✓
c. The search starts over from the beginning
d. The search continues until the end of the list